REQUEST FOR INFORMATION FOR UNDERGRADUATE JET TRAINING SYSTEM (UJTS) FIELDING

APPENDIX A: DRAFT UJTS MISSION PROFILES

Prepared for and by:
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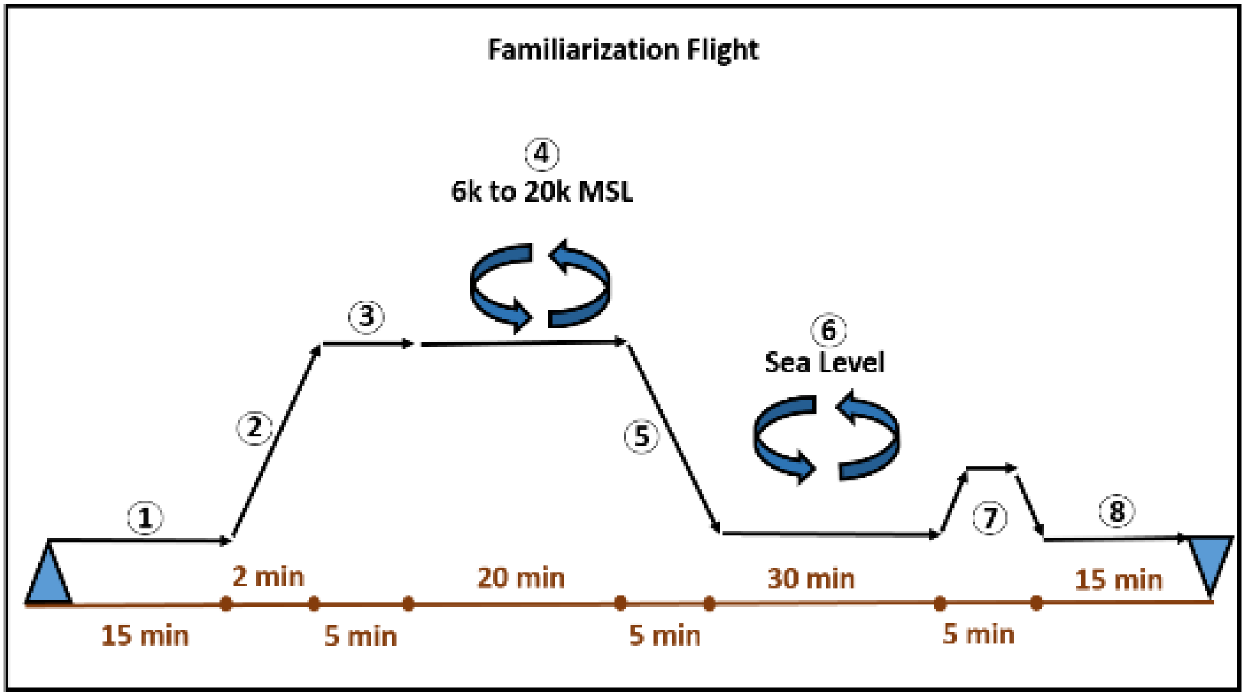
31 Mar 2025

1. Familiarization Flight



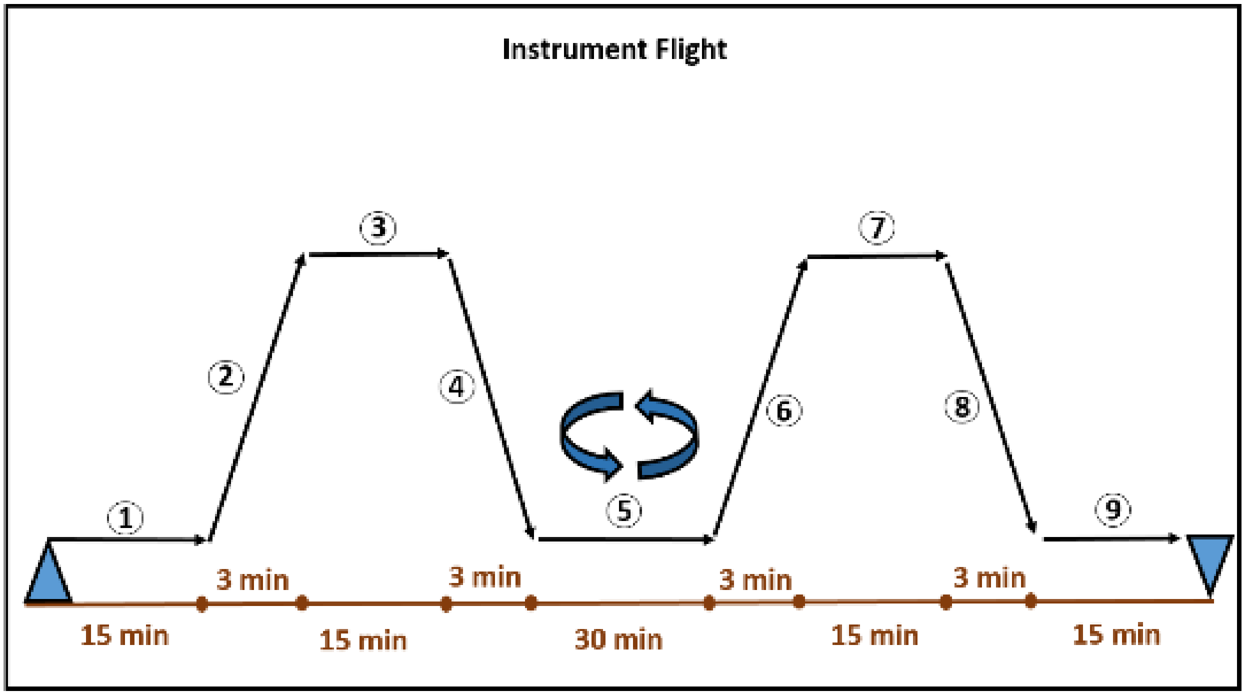
Profile Summary:

- Mission overview: Climb and transit to working area followed by transit / descent to Outlying Field for landing pattern practice then transit home.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (6k feet MSL): 2 minutes at intermediate thrust
3) Transit to working area at 6k feet Mean Sea Level (MSL), cruise at 250 Knots Indicated Airspeed (KIAS): 5 minutes
4) Basic Familiarization (FAM) stage air work (Aerobatics, Stalls, Unusual Attitudes, Slow Flight), calculate below at 6k feet MSL
    a. 6 minutes maximum thrust
    b. 8 minutes at airspeed for max range
    c. 6 minutes at flight idle
5) Enroute descent to Outlying Field (0 feet MSL): 5 minutes at airspeed for max range
6) Six touch and go landings: 30 minutes in landing configuration with on-speed AOA
7) Climb to 3k feet MSL, transit at 3k feet MSL, descent into home field (0 feet MSL): 5 minutes at 200 knots or airspeed for maximum range
8) Full stop landing pattern, followed by taxi and shutdown: 15 minutes

- Total Mission Time: Approximately 75 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

2. Instrument Flight



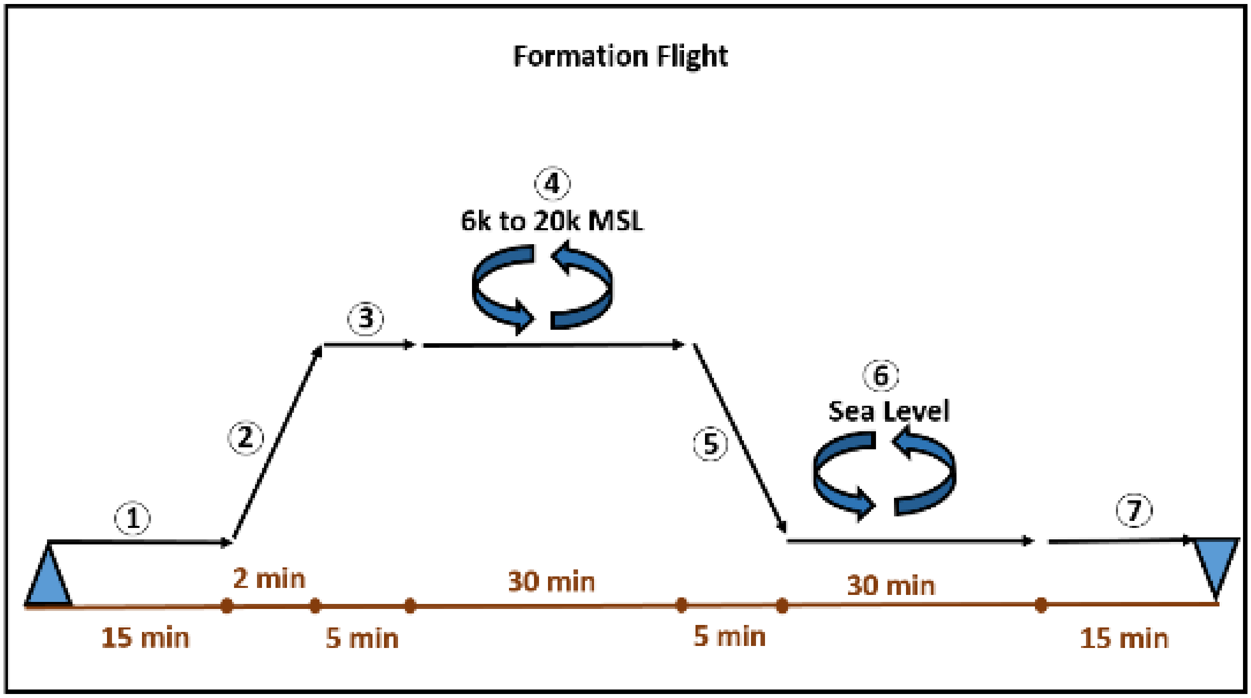
Profile Summary:

- Mission overview: Climb, transit, and enroute descent to an airfield for practice instrument approaches; followed by a return leg to home field for a ground controlled approach and full-stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to altitude for transit (15k feet MSL): 3 minutes at intermediate thrust
3) Transit at 15k feet MSL to enroute descent point at airspeed for maximum range: 15 minutes
4) Enroute descent to sea level at airspeed for max range: 15 minutes
5) Practice instrument approaches: 20 minutes at 200 KIAS at sea level with 10 minutes in landing configuration, on-speed Angle of Attack (AOA) and thrust requirements
6) Climb to altitude for transit (15k feet MSL): 3 minutes at intermediate thrust
7) Transit at 15k feet MSL to enroute descent point at airspeed for maximum range: 15 minutes
8) Enroute descent to Ground Controlled Approach (GCA): 3 minutes at max range descent (250 KIAS to 200 KIAS) followed by 3 minutes at 200 KIAS at sea level for approach power
9) Full-stop landing: 2 minutes at landing configuration on-speed AOA thrust at sea level.

- Total Mission Time: Approximately 80 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

3. Formation Flight



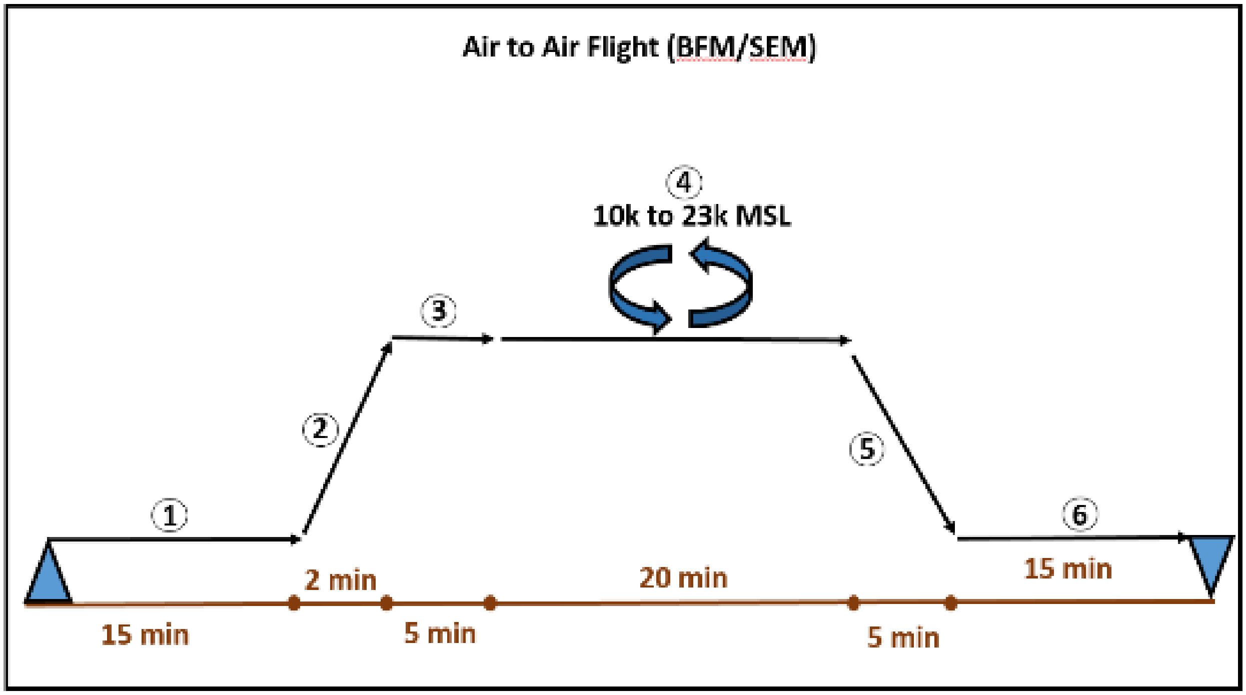
Profile Summary:

- Mission overview: Climb and transit to working area followed by transit/descent to home field for landing pattern practice and full stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (6k feet MSL): 2 minutes at intermediate thrust
3) Transit to working area at 6k feet MSL, cruise at 250 KIAS: 5 minutes
4) Formation stage air work (Break up and Rendezvous, Cruise Maneuvering, Section Approach, Tail-Chase), calculate below at 6k feet MSL
    a. 5 minutes maximum thrust
    b. 15 minutes at airspeed for max range
    c. 10 minutes at intermediate thrust (85-95%)
5) Enroute descent to home field (sea level): 5 minutes at airspeed for max range
6) Six touch and go landings: 30 minutes in landing configuration with on-speed AOA
7) Full stop landing pattern, followed by taxi and shutdown: 15 minutes

- Total Mission Time: Approximately 78 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

4. Air to Air Flight
(Basic Fighter Maneuvers (BFM) / Section Engaged Maneuvers (SEM))



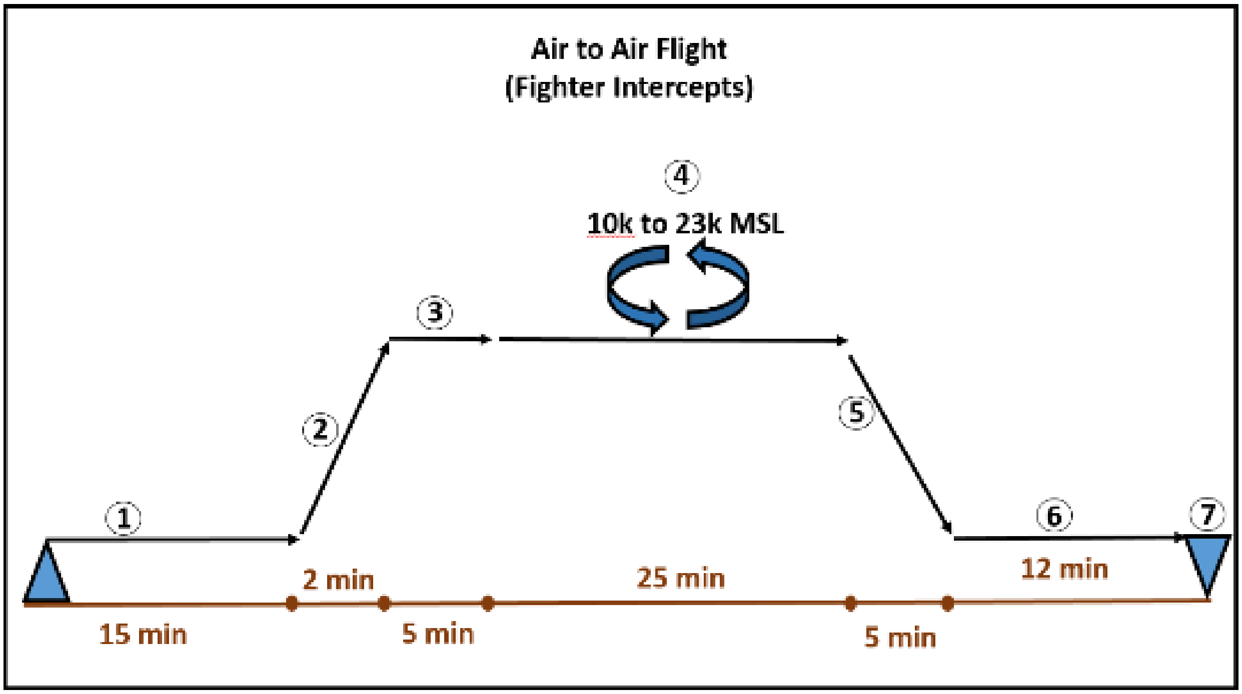
Profile Summary:

- Mission overview: Climb and transit to working area followed by transit/descent to home field for full stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (10k feet MSL): 2 minutes at intermediate thrust
3) Transit to working area at 10k feet MSL, cruise at 300 KIAS: 5 minutes
4) Air-to-Air area work (G-Warm, Offensive Basic Fighter Maneuvers (BFM), Defensive BFM, High-Aspect BFM, Section Engaged Maneuvering), calculate below at 10k feet MSL
    a. 15 minutes maximum thrust
    b. 2 minutes at airspeed for max range
    c. 3 minutes at intermediate thrust (85-95%)
5) Enroute descent to home field (sea level): 5 minutes at airspeed for max range
6) Landing pattern (Practice Powered Approach (PA) and full stop landing): 10 minutes in landing configuration, on-speed AOA required thrust
7) Taxi and shutdown: 5 minutes at idle thrust

- Total Mission Time: Approximately 45 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

5. Air to Air Flight (Fighter Intercepts)



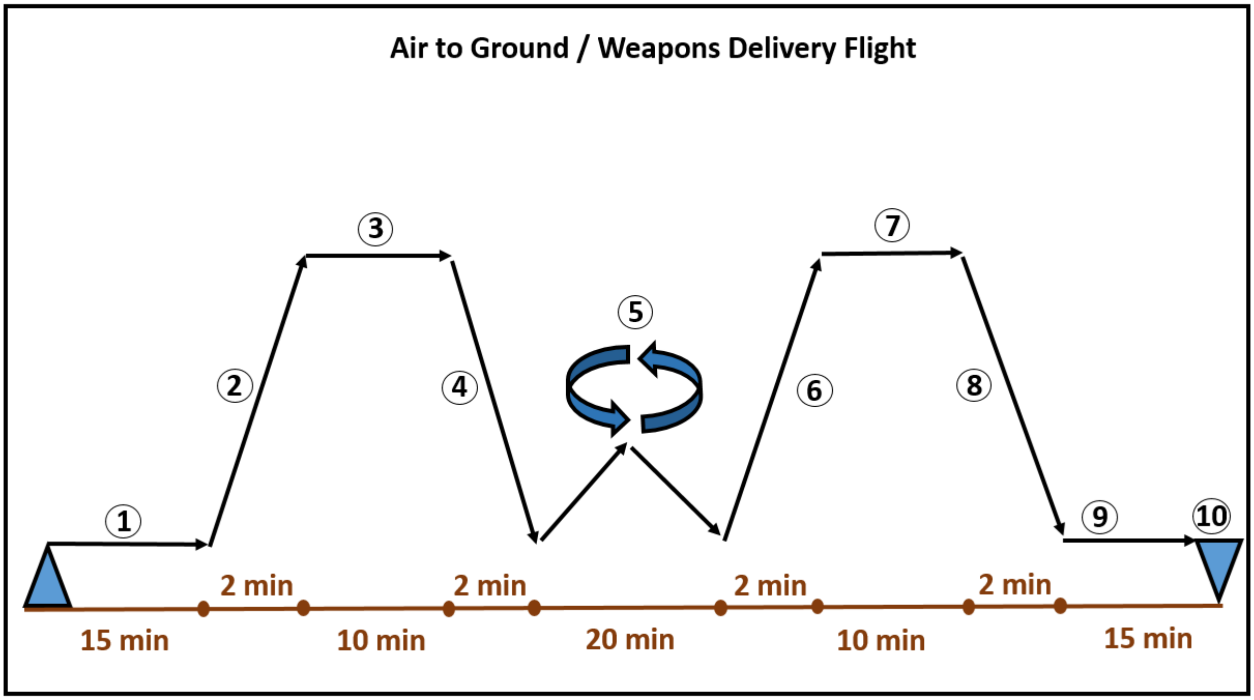
Profile Summary:

- Mission overview: Climb and transit to working area followed by transit/descent to home field for full stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (10k feet MSL): 2 minutes at intermediate thrust
3) Transit to working area, cruise at 300 KIAS: 5 minutes
4) Air-to-Air area work (G-Warm, 3x Hold and run intercept, High-Aspect BFM), calculate below at 10k feet MSL
    a. 18 minutes maximum thrust
    b. 2 minutes at airspeed for max range
    c. 5 minutes at intermediate thrust (85-95%)
5) Enroute descent to home field (sea level): 5 minutes at airspeed for max range
6) Landing pattern (Practice PA and full stop landing): 7 minutes in landing configuration, on-speed AOA required thrust
7) Taxi and shutdown: 5 minutes at idle thrust

- Total Mission Time: Approximately 50 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

6. Air to Ground Flight



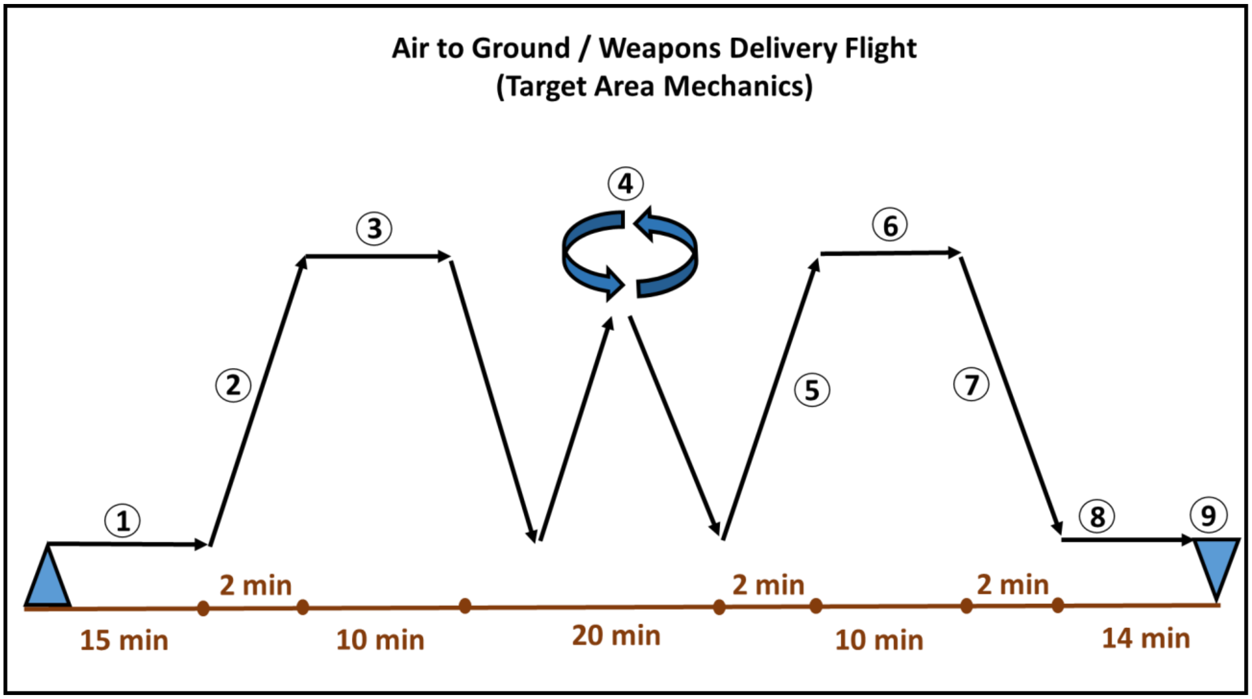
Profile Summary:

- Mission overview: Climb and transit to weapons range followed by transit/descent to home field for straight-in approach or pattern full stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (10k feet MSL): 2 minutes at maximum thrust
3) Transit to working area at 10k feet MSL, cruise at max range airspeed required thrust: 5 minutes
4) Descent into weapons range at 4k feet MSL and "Spacer" pass: Intermediate thrust increasing to maximum thrust for 2 minutes
5) Six to Eight weapons roll-in or pop dive deliveries with pattern: descent to Sea level, climb to 4k feet MSL, and hold at 4k feet MSL for 1 min (20 minutes total)
    a. 15 minutes intermediate thrust (85-95%)
    b. 5 minutes at maximum thrust
6) Rejoin and climb to transit altitude at 5k feet MSL: 2 minutes at intermediate thrust.
7) Transit at 5k feet MSL at max range required thrust: 10 minutes
8) Descent to sea level into landing pattern or straight-in approach: Idle thrust for 2 minutes
9) Landing pattern (Straight-in approach and full stop landing): 8 minutes in landing configuration, on-speed AOA required thrust
10) Taxi and shutdown: 7 minutes at idle thrust

- Total Mission Time: Approximately 57 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

7. Air to Ground Flight (Target Area Mechanics)



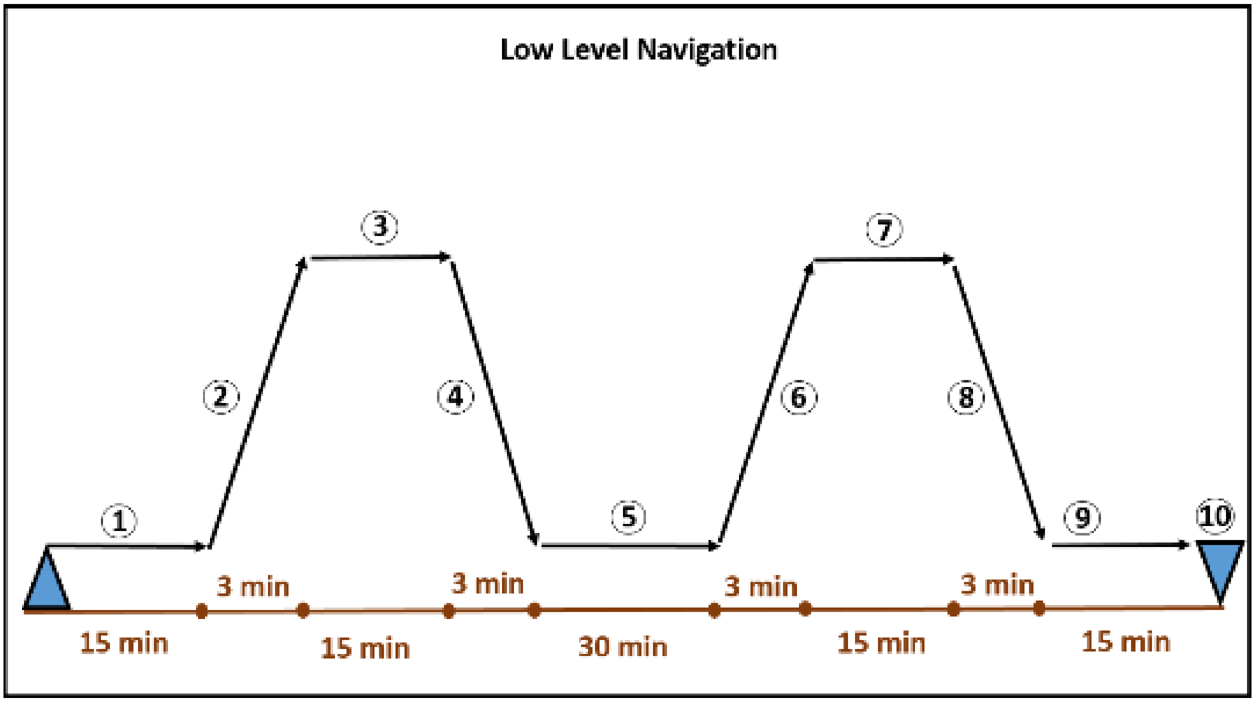
Profile Summary:

- Mission overview: Climb and transit to weapons range followed by transit/descent to home field for straight-in approach or pattern full stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (10k feet MSL): 2 minutes at maximum thrust
3) Transit to working area and hold at Initial Point (IP) at 10k feet MSL, cruise at max range airspeed required thrust: 10 minutes
4) Three to four ingresses from IP to weapons roll-in followed by climb to holding altitude and reset to IP: 10k feet MSL transit, descent to sea level and climb back to 10k MSL, 10k feet MSL transit to IP (20 minutes)
   a. 15 minutes intermediate thrust (85-95%)
   b. 5 minutes at maximum thrust
5) Rejoin and climb to transit altitude (15k feet MSL): 2 minutes at intermediate thrust.
6) Transit at 15k feet MSL at max range required thrust: 10 minutes
7) Descent to sea level into landing pattern or straight-in approach: Idle thrust for 2 minutes
8) Landing pattern (Straight-in approach and full stop landing): 7 minutes in landing configuration, on-speed AOA required thrust
9) Taxi and shutdown: 7 minutes at idle thrust

- Total Mission Time: Approximately 57 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

8. Low Level Navigation



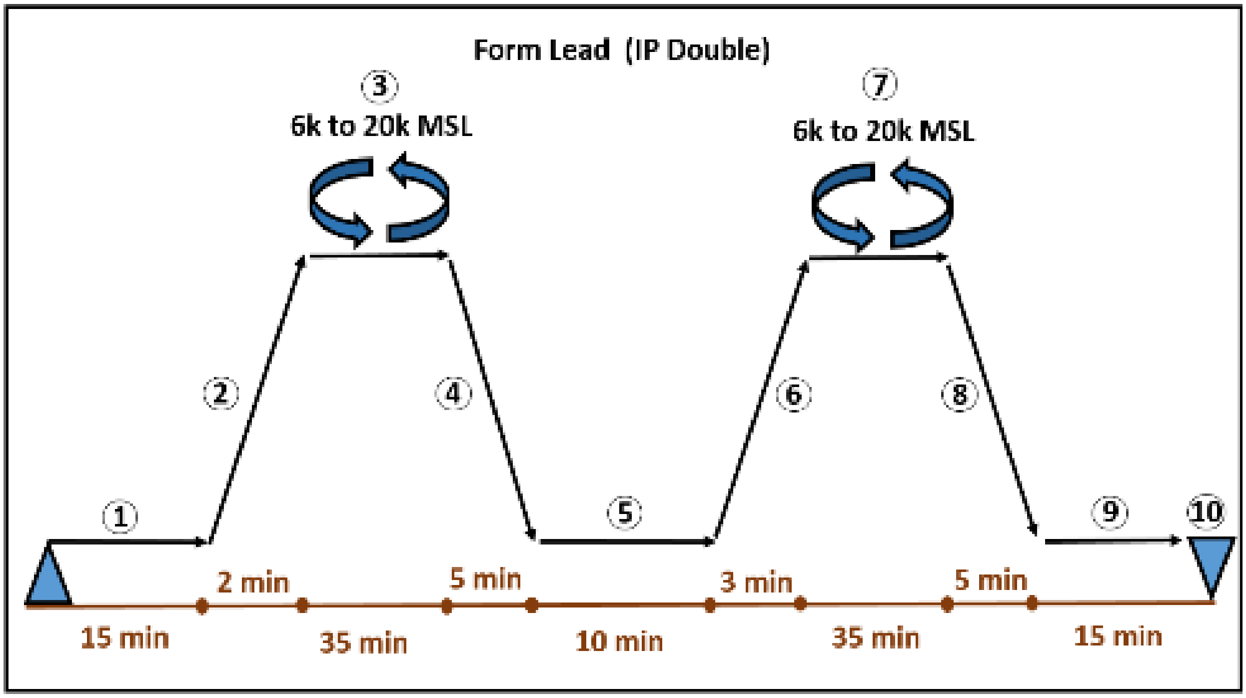
Profile Summary:

- Mission overview: Climb and transit to descent point for entry into low level route structure, followed by climb and transit to airfield for a pattern full stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to transit altitude at 15k feet MSL: 3 minutes at maximum thrust
3) Transit to descent point at 15k feet MSL, cruise at max range airspeed required thrust: 15 minutes
4) Descent into low level route structure (500 feet MSL): Idle thrust for 3 minutes
5) Low Level Navigation at 500 feet MSL: 360 Knots Calibrated Airspeed (KCAS) for max of 30 minutes
6) Climb to 15k MSL at maximum thrust: 3 minutes
7) Transit at 15k MSL at max range required thrust: 10 minutes
8) Descent to sea level into landing pattern: Idle thrust for 3 minutes
9) Landing pattern: 5 minutes in landing configuration, on-speed AOA required thrust
10) Taxi and shutdown: 10 minutes at idle thrust

- Total Mission Time: Approximately 78 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

9. Formation Lead (IP Lead Double Flight with External Fuel)



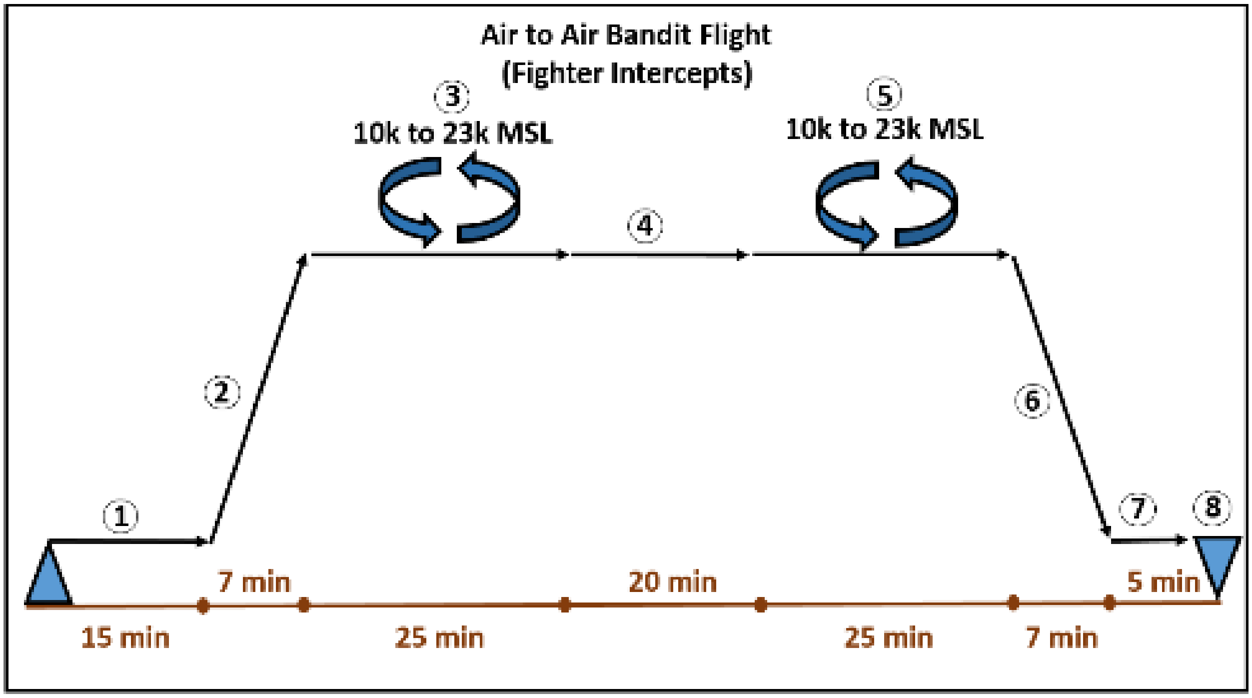
Profile Summary:

- Mission overview: Climb / transit to working area followed by transit/descent to home field; airborne swap of students, repeat event to a full-stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (6k feet MSL): 2 minutes at intermediate thrust
3) Transit to working area, cruise at 6k feet MSL and 250 KIAS: 5 minutes
   Formation stage air work (Break up and Rendezvous, Cruise Maneuvering, Section Approach, Tail-Chase), calculate below at 6k feet MSL
   a. 5 minutes maximum thrust
   b. 15 minutes at airspeed for max range
   c. 10 minutes at intermediate thrust (85-95%)
4) Enroute descent to home field (sea level): 5 minutes at airspeed for max range
5) Section Approach/Missed Approach/Pattern Re-Entry to Section Break: 2 minutes at 200 KIAS at sea level; 3 minutes in landing configuration, on-speed AOA and thrust requirements, 1 minute climb to 3k MSL at 200 KIAS thrust requirement; 3 minute 200 KIAS at 3k MSL with level acceleration to 300 KIAS; 1 minute 300 KIAS thrust requirement, pattern entry into break (idle 300 KIAS to on-speed AOA).
6) Climb to working area altitude for transit (6k feet MSL): 3 minutes at intermediate thrust (includes 1 minute at maximum thrust following Touch-n-Go)
7) Same as #3
8) Same as #4
9) Same as #5
10) Full stop landing pattern, followed by taxi and shutdown: 15 minutes

- Total Mission Time: Approximately 125 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance

10. Air to Air Bandit Flight (IP Double Flight with External Fuel)



Profile Summary:

- Mission overview: Climb and transit to working area, conduct area work followed by max endurance holding while second section of "fighters" enters area. Once complete, conduct transit/descent to home field for full stop landing.

1) Engine start, taxi, and takeoff: 14 minutes at idle thrust, plus 1 minute at maximum thrust (sea level)
2) Climb to working area altitude for transit (10k feet MSL): 2 minutes at intermediate thrust; transit to working area, cruise at 300 KIAS: 5 minutes
3) Air-to-Air area work (G-Warm, 3x Hold at CAP/Run Intercept, High-Aspect BFM), calculate below at 10k feet MSL
    a. 18 minutes maximum thrust
    b. 2 minutes at airspeed for max range
    c. 5 minutes at intermediate thrust (85-95%)
4) Climb to and execute max Endurance holding at 20k MSL, descend to 10k feet MSL: 20 minutes
5) Same as #3
6) Enroute descent to sea level at home field: 7 minutes at airspeed for max range
7) Landing pattern: 5 minutes at on-speed AOA in landing configuration. (Practice PA and full stop landing):
8) Taxi and shutdown: 5 minutes at idle thrust

- Total Mission Time: Approximately 104 minutes (takeoff to landing)
- Reserve fuel: 30 minutes at sea level at airspeed for maximum endurance